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POSE MEASUREMENT CHAINING FOR EXTENDED REALITY SURGICAL NAVIGATION IN VISIBLE AND NEAR INFRARED SPECTRUMS

Abstract

A surgical system includes a camera tracking system that determines a first pose transform between a first object coordinate system and the first tracking camera coordinate system based on first object tracking information from the first tracking camera which indicates pose of the first object. The camera tracking system determines a second pose transform between the first object coordinate system and the second tracking camera coordinate system based on first object tracking information from the second tracking camera indicating pose of the first object, and determines a third pose transform between a second object coordinate system and the second tracking camera coordinate system based on second object tracking information from the second tracking camera indicating pose of the second object. The camera tracking system determines a fourth pose transform between the second object coordinate system and the first tracking camera coordinate system based on combining the first, second, and third pose transforms.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/424,925, filed on Jan. 29, 2024 and published as US 2024-0206986, which is a continuation of U.S. patent application Ser. No. 17/938,860, filed on Oct. 7, 2022, which is a continuation of U.S. patent application Ser. No. 16/774,799, filed on Jan. 28, 2020, each of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to medical devices and systems, and more particularly, computer assisted navigation in surgery.

BACKGROUND

[0003] Computer assisted navigation in surgery provides surgeons with enhanced visualization of surgical instruments with respect to radiographic images of the patient's anatomy. Navigated surgeries typically include components for tracking the position and orientation of surgical instruments via arrays of disks or spheres using a single near infrared (NIR) stereo camera setup. In this scenario, there are three parameters jointly competing for optimization: (1) accuracy, (2) robustness and (3) ergonomics.

[0004] Navigated surgery procedures using existing navigation systems are prone to events triggering intermittent pauses while personnel and/or objects obstruct the ability of a tracking component to track poses of the patient, the robot, and surgical instruments. There is a need to improve the tracking performance of navigation systems.

SUMMARY

[0005] Various embodiments disclosed herein are directed to improvements in computer assisted navigation during surgery. One or more extended reality (XR) headsets can be equipped with tracking cameras that provide tracking information to a camera tracking system for combining with tracking information from other tracking cameras which may be part of another XR headset, an auxiliary tracking bar, or other equipment. Through various pose chaining operations disclosed herein the camera tracking system may be able to track tools and other objects with greater robustness and through a wide range of motion.

[0006] In one embodiment, a surgical system includes a camera tracking system that is configured to receive tracking information related to tracked objects from a first tracking camera and a second tracking camera during a surgical procedure. The camera tracking system is configured to determine a first pose transform between a first object coordinate system and the first tracking camera coordinate system based on first object tracking information from the first tracking camera which indicates pose of the first object. The camera tracking system is configured to determine a second pose transform between the first object coordinate system and the second tracking camera

coordinate system based on first object tracking information from the second tracking camera which indicates pose of the first object, and to determine a third pose transform between a second object coordinate system and the second tracking camera coordinate system based on second object tracking information from the second tracking camera which indicates pose of the second object. The camera tracking system is configured to determine a fourth pose transform between the second object coordinate system and the first tracking camera coordinate system based on combining the first, second, and third pose transforms.

[0007] Related methods by a camera tracking system and related computer program products are disclosed.

[0008] Other surgical systems, method, and computer program products according to embodiments will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such surgical systems, method, and computer program products be included within this description, be within the scope of the present disclosure, and be protected by the accompanying claims. Moreover, it is intended that all embodiments disclosed herein can be implemented separately or combined in any way and/or combination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in a constitute a part of this application, illustrate certain non-limiting embodiments of inventive concepts. In the drawings:

[0010] FIG. 1 illustrates an embodiment of a surgical system according to some embodiments of the present disclosure;

[0011] FIG. 2 illustrates a surgical robot component of the surgical system of FIG. 1 according to some embodiments of the present disclosure;

[0012] FIG. 3A illustrates a camera tracking system component of the surgical system of FIG. 1 according to some embodiments of the present disclosure;

[0013] FIGS. 3B and 3C illustrate a front view and isometric view of another camera tracking system component which may be used with the surgical system of FIG. 1 according to some embodiments of the present disclosure;

[0014] FIG. 4 illustrates an embodiment of an end effector that is connectable to a robot arm and configured according to some embodiments of the present disclosure;

[0015] FIG. 5 illustrates a medical operation in which a surgical robot and a camera system are disposed around a patient;

[0016] FIG. 6 illustrates a block diagram view of the components of the surgical system of FIG. 5 being used for a medical operation;

[0017] FIG. 7 illustrates various display screens that may be displayed on the display of FIGS. 5 and 6 when using a navigation function of the surgical system;

[0018] FIG. 8 illustrates a block diagram of some electrical components of a surgical robot according to some embodiments of the present disclosure;

[0019] FIG. 9 illustrates a block diagram of components of a surgical system that includes imaging devices connected to a computer platform which can be operationally connected to a camera tracking system and/or surgical robot according to some embodiments of the present disclosure;

[0020] FIG. 10 illustrates an embodiment of a C-Arm imaging device that can be used in combination with the surgical robot in accordance with some embodiments of the present disclosure;

[0021] FIG. 11 illustrates an embodiment of an O-Arm imaging device that can be used in combination with the surgical robot in accordance with some embodiments of the present

disclosure;

[0022] FIG. **12** illustrates a block diagram view of the components of a surgical system that includes a pair of XR headsets and an auxiliary tracking bar which operate in accordance with some embodiments of the present disclosure;

[0023] FIG. **13** illustrates an XR headset which is configured in accordance with some embodiments of the present disclosure;

[0024] FIG. **14** illustrates electrical components of the XR headset that can be operatively connected to a computer platform, imaging device(s), and/or a surgical robot in accordance with some embodiments of the present disclosure;

[0025] FIG. **15** illustrates a block diagram showing arrangement of optical components of the XR headset in accordance with some embodiments of the present disclosure;

[0026] FIG. **16** illustrates an example view through the display screen of an XR headset for providing navigation assistance to manipulate a surgical tool during a medical procedure in accordance with some embodiments of the present disclosure;

[0027] FIG. **17** illustrates an example configuration of an auxiliary tracking bar having two pairs of stereo cameras configured in accordance with some embodiments of the present disclosure;

[0028] FIG. **18** illustrates a block diagram view of the components of a surgical system that includes navigation cameras in a pair of XR headsets and in an auxiliary tracking bar which collectively operate in accordance with some embodiments of the present disclosure;

[0029] FIG. **19** illustrates a pair of XR headsets that operate with a camera tracking system to cooperatively tracking pose of dynamic reference array fiducials that cannot be fully directly observed by either XR headset, in accordance with some embodiments of the present disclosure;

[0030] FIG. **20** illustrates a block diagram of information flows between various components of a surgical system operating in accordance with some embodiments of the present disclosure; and

[0031] FIG. **21** is a flowchart of pose chaining operations that can be performed by a camera tracking system in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0032] Inventive concepts will now be described more fully hereinafter with reference to the accompanying drawings, in which examples of embodiments of inventive concepts are shown. Inventive concepts may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of various present inventive concepts to those skilled in the art. It should also be noted that these embodiments are not mutually exclusive. Components from one embodiment may be tacitly assumed to be present or used in another embodiment.

[0033] Various embodiments disclosed herein are directed to improvements in computer assisted navigation during surgery. An extended reality (XR) headset is operatively connected to the surgical system and configured to provide an interactive environment through which a surgeon, assistant, and/or other personnel can view and select among patient images, view and select among computer generated surgery navigation information, and/or control surgical equipment in the operating room. As will be explained below, the XR headset may be configured to augment a real-world scene with computer generated XR images. The XR headset may be configured to provide an augmented reality (AR) viewing environment by displaying the computer generated XR images on a see-through display screen that allows light from the real-world scene to pass therethrough for combined viewing by the user. Alternatively, the XR headset may be configured to provide a virtual reality (VR) viewing environment by preventing or substantially preventing light from the real-world scene from being directly viewed by the user while the user is viewing the computer generated AR images on a display screen. An XR headset can be configured to provide both AR and VR viewing environments. In one embodiment, both AR and VR viewing environments are provided by lateral bands of substantially differing opacity arranged between the see-through

display screen and the real-world scene, so that a VR viewing environment is provided for XR images aligned with a high opacity band and an AR viewing environment is provided for XR images aligned with the low opacity band. In another embodiment, both AR and VR viewing environments are provided by computer adjustable control of an opacity filter that variably constrains how much light from the real-world scene passes through a see-through display screen for combining with the XR images viewed by the user. Thus, the XR headset can also be referred to as an AR headset or a VR headset.

[0034] FIG. 1 illustrates an embodiment of a surgical system 2 according to some embodiments of the present disclosure. Prior to performance of an orthopedic or other surgical procedure, a three-dimensional (“3D”) image scan may be taken of a planned surgical area of a patient using, e.g., the C-Arm imaging device 104 of FIG. 10 or O-Arm imaging device 106 of FIG. 11, or from another medical imaging device such as a computed tomography (CT) image or MRI. This scan can be taken pre-operatively (e.g. few weeks before procedure, most common) or intra-operatively. However, any known 3D or 2D image scan may be used in accordance with various embodiments of the surgical system 2. The image scan is sent to a computer platform in communication with the surgical system 2, such as the computer platform 910 of the surgical system 900 (FIG. 9) which may include the camera tracking system component 6, the surgical robot 4 (e.g., robot 2 in FIG. 1), imaging devices (e.g., C-Arm 104, O-Arm 106, etc.), and an image database 950 for storing image scans of patients. A surgeon reviewing the image scan(s) on a display device of the computer platform 910 (FIG. 9) generates a surgical plan defining a target pose for a surgical tool to be used during a surgical procedure on an anatomical structure of the patient. Example surgical tools, also referred to as tools, can include, without limitation, drills, screw drivers, retractors, and implants such as a screws, spacers, interbody fusion devices, plates, rods, etc. In some embodiments, the surgical plan defining the target plane is planned on the 3D image scan displayed on a display device.

[0035] As used herein, the term “pose” refers to the position and/or the rotational angle of one object (e.g., dynamic reference array, end effector, surgical tool, anatomical structure, etc.) relative to another object and/or to a defined coordinate system. A pose may therefore be defined based on only the multidimensional position of one object relative to another object and/or to a defined coordinate system, only on the multidimensional rotational angles of the object relative to another object and/or to a defined coordinate system, or on a combination of the multidimensional position and the multidimensional rotational angles. The term “pose” therefore is used to refer to position, rotational angle, or combination thereof.

[0036] The surgical system 2 of FIG. 1 can assist surgeons during medical procedures by, for example, holding tools, aligning tools, using tools, guiding tools, and/or positioning tools for use. In some embodiments, surgical system 2 includes a surgical robot 4 and a camera tracking system component 6. The ability to mechanically couple surgical robot 4 and camera tracking system component 6 can allow for surgical system 2 to maneuver and move as a single unit, and allow surgical system 2 to have a small footprint in an area, allow easier movement through narrow passages and around turns, and allow storage within a smaller area.

[0037] A surgical procedure may begin with the surgical system 2 moving from medical storage to a medical procedure room. The surgical system 2 may be maneuvered through doorways, halls, and elevators to reach a medical procedure room. Within the room, the surgical system 2 may be physically separated into two separate and distinct systems, the surgical robot 4 and the camera tracking system component 6. Surgical robot 4 may be positioned adjacent the patient at any suitable location to properly assist medical personnel. Camera tracking system component 6 may be positioned at the base of the patient, at the patient shoulders, or any other location suitable to track the present pose and movement of the pose of tracks portions of the surgical robot 4 and the patient. Surgical robot 4 and camera tracking system component 6 may be powered by an onboard power source and/or plugged into an external wall outlet.

[0038] Surgical robot **4** may be used to assist a surgeon by holding and/or using tools during a medical procedure. To properly utilize and hold tools, surgical robot **4** may rely on a plurality of motors, computers, and/or actuators to function properly. Illustrated in FIG. **1**, robot body **8** may act as the structure in which the plurality of motors, computers, and/or actuators may be secured within surgical robot **4**. Robot body **8** may also provide support for robot telescoping support arm **16**. The size of robot body **8** may provide a solid platform supporting attached components, and may house, conceal, and protect the plurality of motors, computers, and/or actuators that may operate attached components.

[0039] Robot base **10** may act as a lower support for surgical robot **4**. In some embodiments, robot base **10** may support robot body **8** and may attach robot body **8** to a plurality of powered wheels **12**. This attachment to wheels may allow robot body **8** to move in space efficiently. Robot base **10** may run the length and width of robot body **8**. Robot base **10** may be about two inches to about 10 inches tall. Robot base **10** may cover, protect, and support powered wheels **12**.

[0040] In some embodiments, as illustrated in FIG. **1**, at least one powered wheel **12** may be attached to robot base **10**. Powered wheels **12** may attach to robot base **10** at any location. Each individual powered wheel **12** may rotate about a vertical axis in any direction. A motor may be disposed above, within, or adjacent to powered wheel **12**. This motor may allow for surgical system **2** to maneuver into any location and stabilize and/or level surgical system **2**. A rod, located within or adjacent to powered wheel **12**, may be pressed into a surface by the motor. The rod, not pictured, may be made of any suitable metal to lift surgical system **2**. The rod may lift powered wheel **10**, which may lift surgical system **2**, to any height required to level or otherwise fix the orientation of the surgical system **2** in relation to a patient. The weight of surgical system **2**, supported through small contact areas by the rod on each wheel, prevents surgical system **2** from moving during a medical procedure. This rigid positioning may prevent objects and/or people from moving surgical system **2** by accident.

[0041] Moving surgical system **2** may be facilitated using robot railing **14**. Robot railing **14** provides a person with the ability to move surgical system **2** without grasping robot body **8**. As illustrated in FIG. **1**, robot railing **14** may run the length of robot body **8**, shorter than robot body **8**, and/or may run longer the length of robot body **8**. Robot railing **14** may further provide protection to robot body **8**, preventing objects and or personnel from touching, hitting, or bumping into robot body **8**.

[0042] Robot body **8** may provide support for a Selective Compliance Articulated Robot Arm, hereafter referred to as a “SCARA.” A SCARA **24** may be beneficial to use within the surgical system **2** due to the repeatability and compactness of the robotic arm. The compactness of a SCARA may provide additional space within a medical procedure, which may allow medical professionals to perform medical procedures free of excess clutter and confining areas. SCARA **24** may comprise robot telescoping support **16**, robot support arm **18**, and/or robot arm **20**. Robot telescoping support **16** may be disposed along robot body **8**. As illustrated in FIG. **1**, robot telescoping support **16** may provide support for the SCARA **24** and display **34**. In some embodiments, robot telescoping support **16** may extend and contract in a vertical direction. The body of robot telescoping support **16** may be any width and/or height configured to support the stress and weight placed upon it.

[0043] In some embodiments, medical personnel may move SCARA **24** through a command submitted by the medical personnel. The command may originate from input received on display **34**, a tablet, and/or an XR headset (e.g., headset **920** in FIG. **9**) as will be explained in further detail below. The XR headset may eliminate the need for medical personnel to refer to any other display such as the display **34** or a tablet, which enables the SCARA **24** to be configured without the display **34** and/or the tablet. The command may be generated by the depression of a switch and/or the depression of a plurality of switches, and/or may be generated based on a hand gesture command and/or voice command that is sensed by the XR headset as will be explained in further

detail below.

[0044] As shown in FIG. 5, an activation assembly **60** may include a switch and/or a plurality of switches. The activation assembly **60** may be operable to transmit a move command to the SCARA **24** allowing an operator to manually manipulate the SCARA **24**. When the switch, or plurality of switches, is depressed the medical personnel may have the ability to move SCARA **24** through applied hand movements. Alternatively or additionally, an operator may control movement of the SCARA **24** through hand gesture commands and/or voice commands that are sensed by the XR headset as will be explained in further detail below. Additionally, when the SCARA **24** is not receiving a command to move, the SCARA **24** may lock in place to prevent accidental movement by personnel and/or other objects. By locking in place, the SCARA **24** provides a solid platform through which the end effector **26** can guide a surgical tool during a medical procedure.

[0045] Robot support arm **18** can be connected to robot telescoping support **16** by various mechanisms. In some embodiments, best seen in FIGS. 1 and 2, robot support arm **18** rotates in any direction in regard to robot telescoping support **16**. Robot support arm **18** may rotate three hundred and sixty degrees around robot telescoping support **16**. Robot arm **20** may connect to robot support arm **18** at any suitable location and by various mechanisms that enable rotation in any direction relative to robot support arm **18**. In one embodiment, the robot arm **20** can rotate three hundred and sixty degrees relative to the robot support arm **18**. This free rotation allows an operator to position robot arm **20** according to a surgical plan.

[0046] The end effector **26** shown in FIGS. 4 and 5 may attach to robot arm **20** in any suitable location. The end effector **26** can be configured to attach to an end effector coupler **22** of the robot arm **20** positioned by the surgical robot **4**. The example end effector **26** includes a tubular guide that guides movement of an inserted surgical tool relative to an anatomical structure on which a surgical procedure is to be performed.

[0047] In some embodiments, a dynamic reference array **52** is attached to the end effector **26**. Dynamic reference arrays, also referred to as “DRAs” herein, are rigid bodies which may be disposed on an anatomical structure (e.g., bone) of a patient, one or more XR headsets being worn by personnel in the operating room, the end effector, the surgical robot, a surgical tool in a navigated surgical procedure. The computer platform **910** in combination with the camera tracking system component **6** or other 3D localization system are configured to track in real-time the pose (e.g., positions and rotational orientations) of the DRA. The DRA can include fiducials, such as the illustrated arrangement of balls. This tracking of 3D coordinates of the DRA can allow the surgical system **2** to determine the pose of the DRA in any multidimensional space in relation to the target anatomical structure of the patient **50** in FIG. 5.

[0048] As illustrated in FIG. 1, a light indicator **28** may be positioned on top of the SCARA **24**. Light indicator **28** may illuminate as any type of light to indicate “conditions” in which surgical system **2** is currently operating. In some embodiments, the light may be produced by LED bulbs, which may form a ring around light indicator **28**. Light indicator **28** may comprise a fully permeable material that can let light shine through the entirety of light indicator **28**. Light indicator **28** may be attached to lower display support **30**. Lower display support **30**, as illustrated in FIG. 2 may allow an operator to maneuver display **34** to any suitable location. Lower display support **30** may attach to light indicator **28** by any suitable mechanism. In some embodiments, lower display support **30** may rotate about light indicator **28** or be rigidly attached thereto. Upper display support **32** may attach to lower display support **30** by any suitable mechanism.

[0049] In some embodiments, a tablet may be used in conjunction with display **34** and/or without display **34**. The tablet may be disposed on upper display support **32**, in place of display **34**, and may be removable from upper display support **32** during a medical operation. In addition the tablet may communicate with display **34**. The tablet may be able to connect to surgical robot **4** by any suitable wireless and/or wired connection. In some embodiments, the tablet may be able to program and/or control surgical system **2** during a medical operation. When controlling surgical system **2**

with the tablet, all input and output commands may be duplicated on display **34**. The use of a tablet may allow an operator to manipulate surgical robot **4** without having to move around patient **50** and/or to surgical robot **4**.

[0050] As will be explained below, in some embodiments a surgeon and/or other personnel can wear XR headsets that may be used in conjunction with display **34** and/or a tablet or the XR head(s) may eliminate the need for use of the display **34** and/or tablet.

[0051] As illustrated in FIGS. **3A** and **5**, camera tracking system component **6** works in conjunction with surgical robot **4** through wired or wireless communication networks. Referring to FIGS. **1**, **3** and **5**, camera tracking system component **6** can include some similar components to the surgical robot **4**. For example, camera body **36** may provide the functionality found in robot body **8**. Robot body **8** may provide an auxiliary tracking bar upon which cameras **46** are mounted. The structure within robot body **8** may also provide support for the electronics, communication devices, and power supplies used to operate camera tracking system component **6**. Camera body **36** may be made of the same material as robot body **8**. Camera tracking system component **6** may communicate directly to an XR headset, tablet and/or display **34** by a wireless and/or wired network to enable the XR headset, tablet and/or display **34** to control the functions of camera tracking system component **6**.

[0052] Camera body **36** is supported by camera base **38**. Camera base **38** may function as robot base **10**. In the embodiment of FIG. **1**, camera base **38** may be wider than robot base **10**. The width of camera base **38** may allow for camera tracking system component **6** to connect with surgical robot **4**. As illustrated in FIG. **1**, the width of camera base **38** may be large enough to fit outside robot base **10**. When camera tracking system component **6** and surgical robot **4** are connected, the additional width of camera base **38** may allow surgical system **2** additional maneuverability and support for surgical system **2**.

[0053] As with robot base **10**, a plurality of powered wheels **12** may attach to camera base **38**. Powered wheel **12** may allow camera tracking system component **6** to stabilize and level or set fixed orientation in regards to patient **50**, similar to the operation of robot base **10** and powered wheels **12**. This stabilization may prevent camera tracking system component **6** from moving during a medical procedure and may keep cameras **46** on the auxiliary tracking bar from losing track of a DRA connected to an XR headset and/or the surgical robot **4**, and/or losing track of one or more DRAs **52** connected to an anatomical structure **54** and/or tool **58** within a designated area **56** as shown in FIGS. **3A** and **5**. This stability and maintenance of tracking enhances the ability of surgical robot **4** to operate effectively with camera tracking system component **6**. Additionally, the wide camera base **38** may provide additional support to camera tracking system component **6**. Specifically, a wide camera base **38** may prevent camera tracking system component **6** from tipping over when cameras **46** is disposed over a patient, as illustrated in FIGS. **3A** and **5**.

[0054] Camera telescoping support **40** may support cameras **46** on the auxiliary tracking bar. In some embodiments, telescoping support **40** moves cameras **46** higher or lower in the vertical direction. Camera handle **48** may be attached to camera telescoping support **40** at any suitable location and configured to allow an operator to move camera tracking system component **6** into a planned position before a medical operation. In some embodiments, camera handle **48** is used to lower and raise camera telescoping support **40**. Camera handle **48** may perform the raising and lowering of camera telescoping support **40** through the depression of a button, switch, lever, and/or any combination thereof.

[0055] Lower camera support arm **42** may attach to camera telescoping support **40** at any suitable location, in embodiments, as illustrated in FIG. **1**, lower camera support arm **42** may rotate three hundred and sixty degrees around telescoping support **40**. This free rotation may allow an operator to position cameras **46** in any suitable location. Lower camera support arm **42** may connect to telescoping support **40** by any suitable mechanism. Lower camera support arm **42** may be used to provide support for cameras **46**. Cameras **46** may be attached to lower camera support arm **42** by

any suitable mechanism. Cameras **46** may pivot in any direction at the attachment area between cameras **46** and lower camera support arm **42**. In embodiments a curved rail **44** may be disposed on lower camera support arm **42**.

[0056] Curved rail **44** may be disposed at any suitable location on lower camera support arm **42**. As illustrated in FIG. 3A, curved rail **44** may attach to lower camera support arm **42** by any suitable mechanism. Curved rail **44** may be of any suitable shape, a suitable shape may be a crescent, circular, oval, elliptical, and/or any combination thereof. Cameras **46** may be moveably disposed along curved rail **44**. Cameras **46** may attach to curved rail **44** by, for example, rollers, brackets, braces, motors, and/or any combination thereof. Motors and rollers, not illustrated, may be used to move cameras **46** along curved rail **44**. As illustrated in FIG. 3A, during a medical procedure, if an object prevents cameras **46** from viewing one or more DRAs being tracked, the motors may responsively move cameras **46** along curved rail **44**. This motorized movement may allow cameras **46** to move to a new position that is no longer obstructed by the object without moving camera tracking system component **6**. While cameras **46** is obstructed from viewing one or more tracked DRAs, camera tracking system component **6** may send a stop signal to a surgical robot **4**, XR headset, display **34**, and/or a tablet. The stop signal may prevent SCARA **24** from moving until cameras **46** has reacquired tracked DRAs **52** and/or can warn an operator wearing the XR headset and/or viewing the display **34** and/or the tablet. This SCARA **24** can be configured to respond to receipt of a stop signal by stopping further movement of the base and/or end effector coupler **22** until the camera tracking system can resume tracking of DRAs.

[0057] FIGS. 3B and 3C illustrate a front view and isometric view of another camera tracking system component **6'** which may be used with the surgical system of FIG. 1 or may be used independent of a surgical robot. For example, the camera tracking system component **6'** may be used for providing navigated surgery without use of robotic guidance. One of the differences between the camera tracking system component **6'** of FIGS. 3B and 3C and the camera tracking system component **6** of FIG. 3A, is that the camera tracking system component **6'** of FIGS. 3B and 3C includes a housing that transports the computer platform **910**. The computer platform **910** can be configured to perform camera tracking operations to track DRAs, perform navigated surgery operations that provide surgical navigation information to a display device, e.g., XR headset and/or other display device, and perform other computational operations disclosed herein. The computer platform **910** can therefore include a navigation computer, such as one or more of the navigation computers of FIG. 14.

[0058] FIG. 6 illustrates a block diagram view of the components of the surgical system of FIG. 5 used for the medical operation. Referring to FIG. 6, the navigation cameras **46** on the auxiliary tracking bar has a navigation field-of-view **600** in which the pose (e.g., position and orientation) of the reference array **602** attached to the patient, the reference array **604** attached to the surgical instrument, and the robot arm **20** are tracked. The navigation cameras **46** may be part of the camera tracking system component **6'** of FIGS. 3B and 3C, which includes the computer platform **910** configured to perform the operations described below. The reference arrays enable tracking by reflecting light in known patterns, which are decoded to determine their respective poses by the tracking subsystem of the surgical robot **4**. If the line-of-sight between the patient reference array **602** and the navigation cameras **46** in the auxiliary tracking bar is blocked (for example, by a medical personnel, instrument, etc.), further navigation of the surgical instrument may not be able to be performed and a responsive notification may temporarily halt further movement of the robot arm **20** and surgical robot **4**, display a warning on the display **34**, and/or provide an audible warning to medical personnel. The display **34** is accessible to the surgeon **610** and assistant **612** but viewing requires a head to be turned away from the patient and for eye focus to be changed to a different distance and location. The navigation software may be controlled by a tech personnel **614** based on vocal instructions from the surgeon.

[0059] FIG. 7 illustrates various display screens that may be displayed on the display **34** of FIGS. 5

and **6** by the surgical robot **4** when using a navigation function of the surgical system **2**. The display screens can include, without limitation, patient radiographs with overlaid graphical representations of models of instruments that are positioned in the display screens relative to the anatomical structure based on a developed surgical plan and/or based on poses of tracked reference arrays, various user selectable menus for controlling different stages of the surgical procedure and dimension parameters of a virtually projected implant (e.g. length, width, and/or diameter).

[0060] For navigated surgery, various processing components (e.g., computer platform **910**) and associated software described below are provided that enable pre-operatively planning of a surgical procedure, e.g., implant placement, and electronic transfer of the plan to computer platform **910** to provide navigation information to one or more users during the planned surgical procedure.

[0061] For robotic navigation, various processing components (e.g., computer platform **910**) and associated software described below are provided that enable pre-operatively planning of a surgical procedure, e.g., implant placement, and electronic transfer of the plan to the surgical robot **4**. The surgical robot **4** uses the plan to guide the robot arm **20** and connected end effector **26** to provide a target pose for a surgical tool relative to a patient anatomical structure for a step of the planned surgical procedure.

[0062] Various embodiments below are directed to using one or more XR headsets that can be worn by the surgeon **610**, the assistant **612**, and/or other medical personnel to provide an improved user interface for receiving information from and/or providing control commands to the surgical robot, the camera tracking system component **6/6'**, and/or other medical equipment in the operating room.

[0063] FIG. **8** illustrates a block diagram of some electrical components of the surgical robot **4** according to some embodiments of the present disclosure. Referring to FIG. **8**, a load cell (not shown) may be configured to track force applied to end effector coupler **22**. In some embodiments the load cell may communicate with a plurality of motors **850**, **851**, **852**, **853**, and/or **854**. As load cell senses force, information as to the amount of force applied may be distributed from a switch array and/or a plurality of switch arrays to a controller **846**. Controller **846** may take the force information from load cell and process it with a switch algorithm. The switch algorithm is used by the controller **846** to control a motor driver **842**. The motor driver **842** controls operation of one or more of the motors **850**, **851**, **852**, **853**, and **854**. Motor driver **842** may direct a specific motor to produce, for example, an equal amount of force measured by load cell through the motor. In some embodiments, the force produced may come from a plurality of motors, e.g., **850-854**, as directed by controller **846**. Additionally, motor driver **842** may receive input from controller **846**. Controller **846** may receive information from load cell as to the direction of force sensed by load cell.

Controller **846** may process this information using a motion controller algorithm. The algorithm may be used to provide information to specific motor drivers **842**. To replicate the direction of force, controller **846** may activate and/or deactivate certain motor drivers **842**. Controller **846** may control one or more motors, e.g. one or more of **850-854**, to induce motion of end effector **26** in the direction of force sensed by load cell. This force-controlled motion may allow an operator to move SCARA **24** and end effector **26** effortlessly and/or with very little resistance. Movement of end effector **26** can be performed to position end effector **26** in any suitable pose (i.e., location and angular orientation relative to defined three-dimensional (3D) orthogonal reference axes) for use by medical personnel.

[0064] Activation assembly **60**, best illustrated in FIG. **5**, may form of a bracelet that wraps around end effector coupler **22**. The activation assembly **60** may be located on any part of SCARA **24**, any part of end effector coupler **22**, may be worn by medical personnel (and communicate wirelessly), and/or any combination thereof. Activation assembly **60** may comprise of a primary button and a secondary button.

[0065] Depressing primary button may allow an operator to move SCARA **24** and end effector coupler **22**. According to one embodiment, once set in place, SCARA **24** and end effector coupler **22** may not move until an operator programs surgical robot **4** to move SCARA **24** and end effector

coupler **22**, or is moved using primary button. In some examples, it may require the depression of at least two non-adjacent primary activation switches before SCARA **24** and end effector coupler **22** will respond to operator commands. Depression of at least two primary activation switches may prevent the accidental movement of SCARA **24** and end effector coupler **22** during a medical procedure.

[0066] Activated by primary button, load cell may measure the force magnitude and/or direction exerted upon end effector coupler **22** by an operator, i.e. medical personnel. This information may be transferred to one or more motors, e.g. one or more of **850-854**, within SCARA **24** that may be used to move SCARA **24** and end effector coupler **22**. Information as to the magnitude and direction of force measured by load cell may cause the one or more motors, e.g. one or more of **850-854**, to move SCARA **24** and end effector coupler **22** in the same direction as sensed by the load cell. This force-controlled movement may allow the operator to move SCARA **24** and end effector coupler **22** easily and without large amounts of exertion due to the motors moving SCARA **24** and end effector coupler **22** at the same time the operator is moving SCARA **24** and end effector coupler **22**.

[0067] In some examples, a secondary button may be used by an operator as a “selection” device. During a medical operation, surgical robot **4** may notify medical personnel to certain conditions by the XR headset(s) **920**, display **34** and/or light indicator **28**. The XR headset(s) **920** are each configured to display images on a see-through display screen to form an extended reality image that is overlaid on real-world objects viewable through the see-through display screen. Medical personnel may be prompted by surgical robot **4** to select a function, mode, and/or assess the condition of surgical system **2**. Depressing secondary button a single time may activate certain functions, modes, and/or acknowledge information communicated to medical personnel through the XR headset(s) **920**, display **34** and/or light indicator **28**. Additionally, depressing the secondary button multiple times in rapid succession may activate additional functions, modes, and/or select information communicated to medical personnel through the XR headset(s) **920**, display **34** and/or light indicator **28**.

[0068] With further reference to FIG. **8**, electrical components of the surgical robot **4** include platform subsystem **802**, computer subsystem **820**, motion control subsystem **840**, and tracking subsystem **830**. Platform subsystem **802** includes battery **806**, power distribution module **804**, connector panel **808**, and charging station **810**. Computer subsystem **820** includes computer **822**, display **824**, and speaker **826**. Motion control subsystem **840** includes driver circuit **842**, motors **850, 851, 852, 853, 854**, stabilizers **855, 856, 857, 858**, end effector connector **844**, and controller **846**. Tracking subsystem **830** includes position sensor **832** and camera converter **834**. Surgical robot **4** may also include a removable foot pedal **880** and removable tablet computer **890**.

[0069] Input power is supplied to surgical robot **4** via a power source which may be provided to power distribution module **804**. Power distribution module **804** receives input power and is configured to generate different power supply voltages that are provided to other modules, components, and subsystems of surgical robot **4**. Power distribution module **804** may be configured to provide different voltage supplies to connector panel **808**, which may be provided to other components such as computer **822**, display **824**, speaker **826**, driver **842** to, for example, power motors **850-854** and end effector coupler **844**, and provided to camera converter **834** and other components for surgical robot **4**. Power distribution module **804** may also be connected to battery **806**, which serves as temporary power source in the event that power distribution module **804** does not receive power from an input power. At other times, power distribution module **804** may serve to charge battery **806**.

[0070] Connector panel **808** may serve to connect different devices and components to surgical robot **4** and/or associated components and modules. Connector panel **808** may contain one or more ports that receive lines or connections from different components. For example, connector panel **808** may have a ground terminal port that may ground surgical robot **4** to other equipment, a port to

connect foot pedal **880**, a port to connect to tracking subsystem **830**, which may include position sensor **832**, camera converter **834**, and DRA tracking cameras **870**. Connector panel **808** may also include other ports to allow USB, Ethernet, HDMI communications to other components, such as computer **822**. In accordance with some embodiments, the connector panel **808** can include a wired and/or wireless interface for operatively connecting one or more XR headsets **920** to the tracking subsystem **830** and/or the computer subsystem **820**.

[0071] Control panel **816** may provide various buttons or indicators that control operation of surgical robot **4** and/or provide information from surgical robot **4** for observation by an operator. For example, control panel **816** may include buttons to power on or off surgical robot **4**, lift or lower vertical column **16**, and lift or lower stabilizers **855-858** that may be designed to engage casters **12** to lock surgical robot **4** from physically moving. Other buttons may stop surgical robot **4** in the event of an emergency, which may remove all motor power and apply mechanical brakes to stop all motion from occurring. Control panel **816** may also have indicators notifying the operator of certain system conditions such as a line power indicator or status of charge for battery **806**. In accordance with some embodiments, one or more XR headsets **920** may communicate, e.g. via the connector panel **808**, to control operation of the surgical robot **4** and/or to receive and display information generated by surgical robot **4** for observation by persons wearing the XR headsets **920**.

[0072] Computer **822** of computer subsystem **820** includes an operating system and software to operate assigned functions of surgical robot **4**. Computer **822** may receive and process information from other components (for example, tracking subsystem **830**, platform subsystem **802**, and/or motion control subsystem **840**) in order to display information to the operator. Further, computer subsystem **820** may provide output through the speaker **826** for the operator. The speaker may be part of the surgical robot, part of an XR headset **920**, or within another component of the surgical system **2**. The display **824** may correspond to the display **34** shown in FIGS. **1** and **2**.

[0073] Tracking subsystem **830** may include position sensor **832** and camera converter **834**. Tracking subsystem **830** may correspond to the camera tracking system component **6** of FIG. **3**. The DRA tracking cameras **870** operate with the position sensor **832** to determine the pose of DRAs **52**. This tracking may be conducted in a manner consistent with the present disclosure including the use of infrared or visible light technology that tracks the location of active or passive elements of DRAs **52**, such as LEDs or reflective markers, respectively.

[0074] Functional operations of the tracking subsystem **830** and the computer subsystem **820** can be included in the computer platform **910**, which can be transported by the camera tracking system component **6'** of FIGS. **3A** and **3B**. The tracking subsystem **830** can be configured to determine the poses, e.g., location and angular orientation of the tracked DRAs. The computer platform **910** can also include a navigation controller that is configured to use the determined poses to provide navigation information to users that guides their movement of tracked tools relative to position-registered patient images and/or tracked anatomical structures during a planned surgical procedure. The computer platform **910** can display information on the display of FIGS. **3B** and **3C** and/or to one or more XR headsets **920**. The computer platform **910**, when used with a surgical robot, can be configured to communicate with the computer subsystem **820** and other subsystems of FIG. **8** to control movement of the end effector **26**. For example, as will be explained below the computer platform **910** can generate a graphical representation of a patient's anatomical structure, surgical tool, user's hand, etc. with a displayed size, shape, color, and/or pose that is controlled based on the determined pose(s) of one or more the tracked DRAs, and which the graphical representation that is displayed can be dynamically modified to track changes in the determined poses over time.

[0075] Motion control subsystem **840** may be configured to physically move vertical column **16**, upper arm **18**, lower arm **20**, or rotate end effector coupler **22**. The physical movement may be conducted through the use of one or more motors **850-854**. For example, motor **850** may be configured to vertically lift or lower vertical column **16**. Motor **851** may be configured to laterally move upper arm **18** around a point of engagement with vertical column **16** as shown in FIG. **2**.

Motor **852** may be configured to laterally move lower arm **20** around a point of engagement with upper arm **18** as shown in FIG. 2. Motors **853** and **854** may be configured to move end effector coupler **22** to provide translational movement and rotation along in about three-dimensional axes. The computer platform **910** shown in FIG. 9 can provide control input to the controller **846** that guides movement of the end effector coupler **22** to position a passive end effector, which is connected thereto, with a planned pose (i.e., location and angular orientation relative to defined 3D orthogonal reference axes) relative to an anatomical structure that is to be operated on during a planned surgical procedure. Motion control subsystem **840** may be configured to measure position of the end effector coupler **22** and/or the end effector **26** using integrated position sensors (e.g. encoders).

[0076] FIG. 9 illustrates a block diagram of components of a surgical system that includes imaging devices (e.g., C-Arm **104**, O-Arm **106**, etc.) connected to a computer platform **910** which can be operationally connected to a camera tracking system component **6** (FIG. 3A) or **6'** (FIGS. 3B,3C) and/or to surgical robot **4** according to some embodiments of the present disclosure. Alternatively, at least some operations disclosed herein as being performed by the computer platform **910** may additionally or alternatively be performed by components of a surgical system.

[0077] Referring to FIG. 9, the computer platform **910** includes a display **912**, at least one processor circuit **914** (also referred to as a processor for brevity), at least one memory circuit **916** (also referred to as a memory for brevity) containing computer readable program code **918**, and at least one network interface **902** (also referred to as a network interface for brevity). The display **912** may be part of an XR headset **920** in accordance with some embodiments of the present disclosure. The network interface **902** can be configured to connect to a C-Arm imaging device **104** in FIG. 10, an O-Arm imaging device **106** in FIG. 11, another medical imaging device, an image database **950** containing patient medical images, components of the surgical robot **4**, and/or other electronic equipment.

[0078] When used with a surgical robot **4**, the display **912** may correspond to the display **34** of FIG. 2 and/or the tablet **890** of FIG. 8 and/or the XR headset **920** that is operatively connected to the surgical robot **4**, the network interface **902** may correspond to the platform network interface **812** of FIG. 8, and the processor **914** may correspond to the computer **822** of FIG. 8. The network interface **902** of the XR headset **920** may be configured to communicate through a wired network, e.g., thin wire ethernet, and/or through wireless RF transceiver link according to one or more wireless communication protocols, e.g., WLAN, 3GPP 4G and/or 5G (New Radio) cellular communication standards, etc.

[0079] The processor **914** may include one or more data processing circuits, such as a general purpose and/or special purpose processor, e.g., microprocessor and/or digital signal processor. The processor **914** is configured to execute the computer readable program code **918** in the memory **916** to perform operations, which may include some or all of the operations described herein as being performed for surgery planning, navigated surgery, and/or robotic surgery.

[0080] The computer platform **910** can be configured to provide surgery planning functionality. The processor **914** can operate to display on the display device **912** and/or on the XR headset **920** an image of an anatomical structure, e.g., vertebra, that is received from one of the imaging devices **104** and **106** and/or from the image database **950** through the network interface **920**. The processor **914** receives an operator's definition of where the anatomical structure shown in one or more images is to have a surgical procedure, e.g., screw placement, such as by the operator touch selecting locations on the display **912** for planned procedures or using a mouse-based cursor to define locations for planned procedures. When the image is displayed in the XR headset **920**, the XR headset can be configured to sense in gesture-based commands formed by the wearer and/or sense voice based commands spoken by the wearer, which can be used to control selection among menu items and/or control how objects are displayed on the XR headset **920** as will be explained in further detail below.

[0081] The computer platform **910** can be configured to enable anatomy measurement, which can be particularly useful for knee surgery, like measurement of various angles determining center of hip, center of angles, natural landmarks (e.g. transepicondylar line, Whitesides line, posterior condylar line), etc. Some measurements can be automatic while some others can involve human input or assistance. The computer platform **910** may be configured to allow an operator to input a choice of the correct implant for a patient, including choice of size and alignment. The computer platform **910** may be configured to perform automatic or semi-automatic (involving human input) segmentation (image processing) for CT images or other medical images. The surgical plan for a patient may be stored in a cloud-based server, which may correspond to database **950**, for retrieval by the surgical robot **4**.

[0082] During orthopedic surgery, for example, a surgeon may choose which cut to make (e.g. posterior femur, proximal tibia etc.) using a computer screen (e.g. touchscreen) or extended reality (XR) interaction (e.g., hand gesture based commands and/or voice based commands) via, e.g., the XR headset **920**. The computer platform **910** can generate navigation information which provides visual guidance to the surgeon for performing the surgical procedure. When used with the surgical robot **4**, the computer platform **910** can provide guidance that allows the surgical robot **4** to automatically move the end effector **26** to a target pose so that the surgical tool is aligned with a target location to perform the surgical procedure on an anatomical structure.

[0083] In some embodiments, the surgical system **900** can use two DRAs to track patient anatomy position, such as one connected to patient tibia and one connected to patient femur. The system **900** may use standard navigated instruments for the registration and checks (e.g. a pointer similar to the one used in Globus ExcelsiusGPS system for spine surgery).

[0084] A particularly challenging task in navigated surgery is how to plan the position of an implant in spine, knee, and other anatomical structures where surgeons struggle to perform the task on a computer screen which is a 2D representation of the 3D anatomical structure. The system **900** could address this problem by using the XR headset **920** to display a three-dimensional (3D) computer generated representations of the anatomical structure and a candidate implant device. The computer generated representations are scaled and posed relative to each other on the display screen under guidance of the computer platform **910** and which can be manipulated by a surgeon while viewed through the XR headset **920**. A surgeon may, for example, manipulate the displayed computer-generated representations of the anatomical structure, the implant, a surgical tool, etc., using hand gesture based commands and/or voice based commands that are sensed by the XR headset **920**.

[0085] For example, a surgeon can view a displayed virtual handle on a virtual implant, and can manipulate (e.g., grab and move) the virtual handle to move the virtual implant to a desired pose and adjust a planned implant placement relative to a graphical representation of an anatomical structure. Afterward, during surgery, the computer platform **910** could display navigation information through the XR headset **920** that facilitates the surgeon's ability to more accurately follow the surgical plan to insert the implant and/or to perform another surgical procedure on the anatomical structure. When the surgical procedure involves bone removal, the progress of bone removal, e.g., depth of cut, can be displayed in real-time through the XR headset **920**. Other features that may be displayed through the XR headset **920** can include, without limitation, gap or ligament balance along a range of joint motion, contact line on the implant along the range of joint motion, ligament tension and/or laxity through color or other graphical renderings, etc.

[0086] The computer platform **910**, in some embodiments, can allow planning for use of standard surgical tools and/or implants, e.g., posterior stabilized implants and cruciate retaining implants, cemented and cementless implants, revision systems for surgeries related to, for example, total or partial knee and/or hip replacement and/or trauma.

[0087] An automated imaging system can be used in conjunction with the computer platform **910** to acquire pre-operative, intra-operative, post-operative, and/or real-time image data of an

anatomical structure. Example automated imaging systems are illustrated in FIGS. **10** and **11**. In some embodiments, the automated imaging system is a C-arm **104** (FIG. **10**) imaging device or an O-arm® **106** (FIG. **11**). (O-arm® is copyrighted by Medtronic Navigation, Inc. having a place of business in Louisville, Colo., USA). It may be desirable to take x-rays of a patient from a number of different positions, without the need for frequent manual repositioning of the patient which may be required in an x-ray system. C-arm **104** x-ray diagnostic equipment may solve the problems of frequent manual repositioning and may be well known in the medical art of surgical and other interventional procedures. As illustrated in FIG. **10**, a C-arm includes an elongated C-shaped member terminating in opposing distal ends **112** of the “C” shape. C-shaped member is attached to an x-ray source **114** and an image receptor **116**. The space within C-arm **104** of the arm provides room for the physician to attend to the patient substantially free of interference from the x-ray support structure.

[0088] The C-arm is mounted to enable rotational movement of the arm in two degrees of freedom, (i.e. about two perpendicular axes in a spherical motion). C-arm is slidably mounted to an x-ray support structure, which allows orbiting rotational movement of the C-arm about its center of curvature, which may permit selective orientation of x-ray source **114** and image receptor **116** vertically and/or horizontally. The C-arm may also be laterally rotatable, (i.e. in a perpendicular direction relative to the orbiting direction to enable selectively adjustable positioning of x-ray source **114** and image receptor **116** relative to both the width and length of the patient). Spherically rotational aspects of the C-arm apparatus allow physicians to take x-rays of the patient at an optimal angle as determined with respect to the particular anatomical condition being imaged.

[0089] The O-arm® **106** illustrated in FIG. **11** includes a gantry housing **124** which may enclose an image capturing portion, not illustrated. The image capturing portion includes an x-ray source and/or emission portion and an x-ray receiving and/or image receiving portion, which may be disposed about one hundred and eighty degrees from each other and mounted on a rotor (not illustrated) relative to a track of the image capturing portion. The image capturing portion may be operable to rotate three hundred and sixty degrees during image acquisition. The image capturing portion may rotate around a central point and/or axis, allowing image data of the patient to be acquired from multiple directions or in multiple planes.

[0090] The O-arm® **106** with the gantry housing **124** has a central opening for positioning around an object to be imaged, a source of radiation that is rotatable around the interior of gantry housing **124**, which may be adapted to project radiation from a plurality of different projection angles. A detector system is adapted to detect the radiation at each projection angle to acquire object images from multiple projection planes in a quasi-simultaneous manner. The gantry may be attached to a support structure O-arm® support structure, such as a wheeled mobile cart with wheels, in a cantilevered fashion. A positioning unit translates and/or tilts the gantry to a planned position and orientation, preferably under control of a computerized motion control system. The gantry may include a source and detector disposed opposite one another on the gantry. The source and detector may be secured to a motorized rotor, which may rotate the source and detector around the interior of the gantry in coordination with one another. The source may be pulsed at multiple positions and orientations over a partial and/or full three hundred and sixty degree rotation for multi-planar imaging of a targeted object located inside the gantry. The gantry may further comprise a rail and bearing system for guiding the rotor as it rotates, which may carry the source and detector. Both and/or either O-arm® **106** and C-arm **104** may be used as automated imaging system to scan a patient and send information to the surgical system **2**.

[0091] Images captured by an imaging system can be displayed on the XR headset **920** and/or another display device of the computer platform **910**, the surgical robot **4**, and/or another component of the surgical system **900**. The XR headset **920** may be connected to one or more of the imaging devices **104** and/or **106** and/or to the image database **950**, e.g., via the computer platform **910**, to display images therefrom. A user may provide control inputs through the XR

headset **920**, e.g., gesture and/or voice based commands, to control operation of one or more of the imaging devices **104** and/or **106** and/or the image database **950**.

[0092] FIG. **12** illustrates a block diagram view of the components of a surgical system that include a pair of XR headsets **1200** and **1210** (head-mounted displays HMD1 and HMD2), which may correspond to the XR headset **920** shown in FIG. **13** and operate in accordance with some embodiments of the present disclosure.

[0093] Referring to the example scenario of FIG. **12**, the assistant **612** and surgeon **610** are both wearing the XR headsets **1210** and **1210**, respectively. It is optional for the assistant **612** to wear the XR headset **1210**. The XR headsets **1200** and **1210** are configured to provide an interactive environment through which the wearers can view and interact with information related to a surgical procedure as will be described further below. This interactive XR based environment may eliminate a need for the tech personnel **614** to be present in the operating room and may eliminate a need for use of the display **34** shown in FIG. **6**. Each XR headset **1200** and **1210** can include one or more cameras that are be configured to provide an additional source of tracking of DRAs or other reference arrays attached to instruments, an anatomical structure, the end effector **26**, and/or other equipment. In the example of FIG. **12**, XR headset **1200** has a field-of-view (FOV) **1202** for tracking DRAs and other objects, XR headset **1210** has a FOV **1212** partially overlapping FOV **1202** for tracking DRAs and other objects, and the navigation cameras **46** has another FOV **600** partially overlapping FOVs **1202** and **1212** for tracking DRAs and other objects.

[0094] If one or more cameras is obstructed from viewing a DRA attached to a tracked object, e.g., a surgical instrument, but the DRA is in view of one or more other cameras the tracking subsystem **830** and/or navigation controller **828** can continue to track the object seamlessly without loss of navigation. Additionally, if there is partial occlusion of the DRA from the perspective of one camera, but the entire DRA is visible via multiple camera sources, the tracking inputs of the cameras can be merged to continue navigation of the DRA. One of the XR headsets and/or the navigation cameras **46** may view and track the DRA on another one of the XR headsets to enable the computer platform **910** (FIGS. **9** and **14**), the tracking subsystem **830**, and/or another computing component to determine the pose of the DRA relative to one or more defined coordinate systems, e.g., of the XR headsets **1200/1210**, the navigation cameras **46**, and/or another coordinate system defined for the patient, table, and/or room.

[0095] The XR headsets **1200** and **1210** can be operatively connected to view video, pictures, and/or other information received from and/or to provide commands that control various equipment in the surgical room, including but not limited to neuromonitoring, microscopes, video cameras, and anesthesia systems. Data from the various equipment may be processed and displayed within the headset, for example the display of patient vitals or the microscope feed.

Example XR Headset Components and Integration to Navigated Surgery, Surgical Robots, and Other Equipment

[0096] FIG. **13** illustrates an XR headset **920** which is configured in accordance with some embodiments of the present disclosure. The XR headset includes a headband **1306** configured to secure the XR headset to a wearer's head, an electronic component enclosure **1304** supported by the headband **1306**, and a display screen **1302** that extends laterally across and downward from the electronic component enclosure **1304**. The display screen **1302** may be a see-through LCD display device or a semi-reflective lens that reflects images projected by a display device toward the wearer's eyes. A set of DRA fiducials, e.g., dots are painted or attached in a spaced apart known arranged on one or both sides of the headset. The DRA on the headset enables the navigation cameras on the auxiliary tracking bar to track pose of the headset **920** and/or enables another XR headset to track pose of the headset **920**.

[0097] The display screen **1302** operates as a see-through display screen, also referred to as a combiner, that reflects light from display panels of a display device toward the user's eyes. The display panels can be located between the electronic component enclosure and the user's head, and

angled to project virtual content toward the display screen **1302** for reflection toward the user's eyes. The display screen **1302** is semi-transparent and semi-reflective allowing the user to see reflected virtual content superimposed on the user's view of a real-world scene. The display screen **1302** may have different opacity regions, such as the illustrated upper laterally band which has a higher opacity than the lower laterally band. Opacity of the display screen **1302** may be electronically controlled to regulate how much light from the real-world scene passes through to the user's eyes. A high opacity configuration of the display screen **1302** results in high-contrast virtual images overlaid on a dim view of the real-world scene. A low opacity configuration of the display screen **1302** can result in more faint virtual images overlaid on a clearer view of the real-world scene. The opacity may be controlled by applying an opaque material on a surface of the display screen **1302**.

[0098] According to some embodiments the surgical system includes an XR headset **920** and an XR headset controller, e.g., controller **1430** in FIG. **14** or controller **3410** in FIG. **34**. The XR headset **920** is configured to be worn by a user during a surgical procedure and has a see-through display screen **1302** that is configured to display an XR image and to allow at least a portion of a real-world scene to pass therethrough for viewing by the user. The XR headset **920** also includes an opacity filter positioned between at least one of the user's eyes and the real-world scene when the see-through display screen **1302** is viewed by the user. The opacity filter is configured to provide opaqueness to light from the real-world scene. The XR headset controller is configured to communicate with a navigation controller, e.g., controller(s) **828A**, **828B**, and/or **828C** in FIG. **14**, to receive navigation information from the navigation controller which provides guidance to the user during the surgical procedure on an anatomical structure, and is further configured to generate the XR image based on the navigation information for display on the see-through display screen **1302**.

[0099] Opacity of the display screen **1302** may be configured as a gradient having a more continuously changing opacity with distance downward from a top portion of the display screen **1302**. The gradient's darkest point can be located at the top portion of the display screen **1302**, and gradually becoming less opaque further down on the display screen **1302** until the opacity is transparent or not present. In an example further embodiment, the gradient can change from about 90% opacity to entirely transparent approximately at the mid-eye level of the display screen **1302**. With the headset properly calibrated and positioned, the mid-eye level can correspond to the point where the user would look straight out, and the end of the gradient would be located at the "horizon" line of the eye. The darker portion of the gradient will allow crisp, clear visuals of the virtual content and help to block the intrusive brightness of the overhead operating room lights.

[0100] Using an opacity filter in this manner enables the XR headset **920** to provide virtual reality (VR) capabilities, by substantially or entirely blocking light from the real-world scene, along an upper portion of the display screen **1302** and to provide AR capabilities along a middle or lower portion of the display screen **1302**. This allows the user to have the semi-translucence of AR where needed and allowing clear optics of the patient anatomy during procedures. Configuring the display screen **1302** as a gradient instead of as a more constant opacity band can enable the wearer to experience a more natural transition between a more VR type view to a more AR type view without experiencing abrupt changes in brightness of the real-world scene and depth of view that may otherwise strain the eyes such as during more rapid shifting between upward and downward views.

[0101] The display panels and display screen **1302** can be configured to provide a wide field of view see-through XR display system. In one example configuration they provide an 80° diagonal field-of-view (FOV) with 55° of vertical coverage for a user to view virtual content. Other diagonal FOV angles and vertical coverage angles can be provided through different size display panels, different curvature lens, and/or different distances and angular orientations between the display panels and curved display screen **1302**.

[0102] FIG. **14** illustrates electrical components of the XR headset **920** that can be operatively

connected to the computer platform **910**, to one or more of the imaging devices, such as the C-arm imaging device **104**, the O-arm imaging device **106**, and/or the image database **950**, and/or to the surgical robot **800** in accordance with various embodiments of the present disclosure.

[0103] The XR headset **920** provides an improved human interface for performing navigated surgical procedures. The XR headset **920** can be configured to provide functionalities, e.g., via the computer platform **910**, that include without limitation any one or more of: identification of hand gesture based commands and/or voice based commands, display XR graphical objects on a display device **1450**. The display device **1450** may be a video projector, flat panel display, etc., which projects the displayed XR graphical objects on the display screen **1302**. The user can view the XR graphical objects as an overlay anchored to particular real-world objects viewed through the display screen **1302** (FIG. 13). The XR headset **920** may additionally or alternatively be configured to display on the display screen **1450** video feeds from cameras mounted to one or more XR headsets **920** and other cameras.

[0104] Electrical components of the XR headset **920** can include a plurality of cameras **1440**, a microphone **1442**, a gesture sensor **1444**, a pose sensor (e.g., inertial measurement unit (IMU)) **1446**, a display module **1448** containing the display device **1450**, and a wireless/wired communication interface **1452**. As will be explained below, the cameras **1440** of the XR headset may be visible light capturing cameras, near infrared capturing cameras, or a combination of both.

[0105] The cameras **1440** may be configured to operate as the gesture sensor **1444** by capturing for identification user hand gestures performed within the field of view of the camera(s) **1440**. Alternatively the gesture sensor **1444** may be a proximity sensor and/or a touch sensor that senses hand gestures performed proximately to the gesture sensor **1444** and/or senses physical contact, e.g., tapping on the sensor or the enclosure **1304**. The pose sensor **1446**, e.g., IMU, may include a multi-axis accelerometer, a tilt sensor, and/or another sensor that can sense rotation and/or acceleration of the XR headset **920** along one or more defined coordinate axes. Some or all of these electrical components may be contained in the component enclosure **1304** or may be contained in another enclosure configured to be worn elsewhere, such as on the hip or shoulder.

[0106] As explained above, the surgical system **2** includes a camera tracking system component **6/6'** and a tracking subsystem **830** which may be part of the computer platform **910**. The surgical system may include imaging devices (e.g., C-arm **104**, O-arm **106**, and/or image database **950**) and/or a surgical robot **4**. The tracking subsystem **830** is configured to determine a pose of DRAs attached to an anatomical structure, an end effector, a surgical tool, etc. A navigation controller **828** is configured to determine a target pose for the surgical tool relative to an anatomical structure based on a surgical plan, e.g., from a surgical planning function performed by the computer platform **910** of FIG. 9, defining where a surgical procedure is to be performed using the surgical tool on the anatomical structure and based on a pose of the anatomical structure determined by the tracking subsystem **830**. The navigation controller **828** may be further configured to generate steering information based on the target pose for the surgical tool, the pose of the anatomical structure, and the pose of the surgical tool and/or the end effector, where the steering information indicates where the surgical tool and/or the end effector of a surgical robot should be moved to perform the surgical plan.

[0107] The electrical components of the XR headset **920** can be operatively connected to the electrical components of the computer platform **910** through a wired/wireless interface **1452**. The electrical components of the XR headset **920** may be operatively connected, e.g., through the computer platform **910** or directly connected, to various imaging devices, e.g., the C-arm imaging device **104**, the I/O-arm imaging device **106**, the image database **950**, and/or to other medical equipment through the wired/wireless interface **1452**.

[0108] The surgical system **2** further includes at least one XR headset controller **1430** (also referred to as “XR headset controller” for brevity) that may reside in the XR headset **920**, the computer platform **910**, and/or in another system component connected via wired cables and/or wireless

communication links. Various functionality is provided by software executed by the XR headset controller **1430**. The XR headset controller **1430** is configured to receive navigation information from the navigation controller **828** which provides guidance to the user during the surgical procedure on an anatomical structure, and is configured to generate an XR image based on the navigation information for display on the display device **1450** for projection on the see-through display screen **1302**.

[0109] The configuration of the display device **1450** relative to the display screen (also referred to as “see-through display screen”) **1302** is configured to display XR images in a manner such that when the user wearing the XR headset **920** looks through the display screen **1302** the XR images appear to be in the real world. The display screen **1302** can be positioned by the headband **1306** in front of the user's eyes.

[0110] The XR headset controller **1430** can be within a housing that is configured to be worn on a user's head or elsewhere on the user's body while viewing the display screen **1302** or may be remotely located from the user viewing the display screen **1302** while being communicatively connected to the display screen **1302**. The XR headset controller **1430** can be configured to operationally process signaling from the cameras **1440**, the microphone **142**, and/or the pose sensor **1446**, and is connected to display XR images on the display device **1450** for user viewing on the display screen **1302**. Thus, the XR headset controller **1430** illustrated as a circuit block within the XR headset **920** is to be understood as being operationally connected to other illustrated components of the XR headset **920** but not necessarily residing within a common housing (e.g., the electronic component enclosure **1304** of FIG. 13) or being otherwise transportable by the user. For example, the XR headset controller **1430** may reside within the computer platform **910** which, in turn, may reside within a housing of the computer tracking system component **6'** shown in FIGS. 3B and 3C.

Example XR Headset Component Optical Arrangement

[0111] FIG. 34 illustrates a block diagram showing arrangement of optical components of the XR headset **920** in accordance with some embodiments of the present disclosure. Referring to FIG. 34, the display device **1450** is configured to display XR images generated by the XR headset controller **1430**, light from which is projected as XR images **1450** toward the display screen **1302**. The display screen **1302** is configured to combine light of the XR images **1450** and light from the real-world scene **1502** into a combined augmented view **1504** that is directed to the user's eye(s) **1510**. The display screen **1302** configured in this manner operates as a see-through display screen. The XR headset **920** can include any plural number of navigation cameras **1440**. The cameras **1440** may be visible light capturing cameras, near infrared capturing cameras, or a combination of both.

Example User Views Through the XR Headset

[0112] The XR headset operations can display both 2D images and 3D models on the display screen **1302**. The 2D images may preferably be displayed in a more opaque band of the display screen **1302** (upper band) and the 3D model may be more preferably displayed in the more transparent band of the display screen **1302**, otherwise known as the environmental region (bottom band). Below the lower band where the display screen **1302** ends the wearer has an unobstructed view of the surgical room. It is noted that where XR content is displayed on the display screen **1302** may be fluidic. It is possible that where the 3D content is displayed moves to the opaque band depending on the position of the headset relative to the content, and where 2D content is displayed can be placed in the transparent band and stabilized to the real world. Additionally, the entire display screen **1302** may be darkened under electronic control to convert the headset into virtual reality for surgical planning or completely transparent during the medical procedure. As explained above, the XR headset **920** and associated operations not only support navigated procedures, but also can be performed in conjunction with robotically assisted procedures.

[0113] FIG. 16 illustrates an example view through the display screen **1302** of the XR headset **920** for providing navigation assistance to a user who is manipulating a surgical tool **1602** during a

medical procedure in accordance with some embodiments of the present disclosure. Referring to FIG. 16, when the surgical tool **1602** is brought in vicinity of a tracked anatomical structure so that dynamic reference arrays **1630** and **1632**, connected to the surgical tool **1602**, become within the field of view of the cameras **1440** (FIG. 15) and/or **46** (FIG. 6), a graphical representation **1600** of the tool can be displayed in 2D and/or 3D images in relation to a graphical representation **1610** of the anatomical structure. The user can use the viewed graphical representations to adjust a trajectory **1620** of the surgical tool **1602**, which can be illustrated as extending from the graphical representation **2000** of the tool through the graphical representation **1610** of the anatomical structure. The XR headset **920** may also display textual information and other objects **1640**. The dashed line **1650** extending across the viewed display screen represents an example division between different opacity level upper and lower bands.

[0114] Other types of XR images (virtual content) that can be displayed on the display screen **1302** can include, but are not limited to any one or more of: [0115] 1) 2D Axial, Sagittal and/or Coronal views of patient anatomy; [0116] 2) overlay of planned vs currently tracked tool and surgical implant locations; [0117] 3) gallery of preoperative images; [0118] 4) video feeds from microscopes and other similar systems or remote video conferencing; [0119] 5) options and configuration settings and buttons; [0120] 6) floating 3D models of patient anatomy with surgical planning information; [0121] 7) real-time tracking of surgical instruments relative to floating patient anatomy; [0122] 8) augmented overlay of patient anatomy with instructions and guidance; and [0123] 9) augmented overlay of surgical equipment.

Example Configuration of Cameras for Tracking System Component

[0124] FIG. 17 illustrates example configuration of an auxiliary tracking bar **46** having two pairs of stereo navigation cameras configured in accordance with some embodiments of the present disclosure. The auxiliary tracking bar **46** is part of the camera tracking system component of FIGS. 3A, 3B, and 3C. The stereo navigation cameras include a stereo pair of spaced apart visible light capturing cameras and another stereo pair of spaced apart near infrared capturing cameras, in accordance with one embodiment. Alternatively, only one stereo pair of visible light capturing cameras or only one stereo pair of near infrared capture cameras can be used in the auxiliary tracking bar **46**. Any plural number of near infrared and/or visible light cameras can be used.

Pose Measurement Chaining

[0125] As explained above, navigated surgery can include computer vision tracking and determination of pose (e.g., position and orientation in a six degree-of-freedom coordinate system) of surgical instruments, such as by determining pose of attached DRAs that include spaced apart fiducials, e.g., disks or spheres, arranged in a known manner. The computer vision uses spaced apart navigation cameras, e.g., stereo cameras, that are configured to capture near infrared and/or visible light. In this scenario, there are three parameters jointly competing for optimization: (1) accuracy, (2) robustness, and (3) user ergonomics during a surgical procedure.

[0126] Some further aspects of the present disclosure are directed to computer operations that combine (chain) measured poses in ways that can improve optimization of one or more of the above three parameters by incorporating additional navigation cameras mounted to one or more XR headsets. As shown in FIG. 17, a stereo pair of visible light tracking cameras and another stereo pair of near infrared tracking navigation cameras can be attached to the auxiliary tracking bar of the camera tracking system component in accordance with some embodiments of the present disclosure. Operational algorithms are disclosed that analyze the pose of DRAs that are fully observed or partially observed (e.g., when less than all of the fiducials of a DRA are viewed by a pair of stereo cameras), and combine the observed poses or partial poses in ways that can improve accuracy, robustness, and/or ergonomics during navigated surgery.

[0127] As explained above, the XR headset may be configured to augment a real-world scene with computer generated XR images. The XR headset may be configured to provide an XR viewing environment by displaying the computer generated XR images on a see-through display screen that

allows light from the real-world scene to pass therethrough for combined viewing by the user. Alternatively, the XR headset may be configured to provide a VR viewing environment by preventing or substantially preventing light from the real-world scene from being directly viewed by the user along the viewing path of the displayed XR images. An XR headset can be configured to provide both AR and VR viewing environments. In one embodiment, both AR and VR viewing environments are provided by lateral bands of substantially differing opacity arranged between the see-through display screen and the real-world scene, so that a VR viewing environment is provided for XR images aligned with a high opacity band and an AR viewing environment is provided for XR images aligned with the low opacity band. In another embodiment, both AR and VR viewing environments are provided by computer adjustable control of an opacity filter that variably constrains how much light from the real-world scene passes through a see-through display screen for combining with the XR images viewed by the user. Thus, the XR headset can also be referred to as an AR headset or a VR headset.

[0128] As was also explained above, the XR headset can include near infrared tracking cameras and/or visible light tracking cameras that are configured to track fiducials of DRAs connected to surgical instruments, patient anatomy, other XR headset(s), and/or a robotic end effector. Using near infrared tracking and/or visible light tracking on the XR headset provides additional tracking volume coverage beyond what cameras on a single auxiliary tracking bar can provide. Adding near infrared tracking cameras to the existing auxiliary tracking bar allows for the headset location to be tracked more robustly but less accurately than in visible light. Mechanically calibrating the visible and near infrared tracking coordinate systems enables the coordinate systems to be aligned sufficiently to perform 3D DRA fiducials triangulation operations using stereo matching to jointly identify pose of the DRA fiducials between the visible and near infrared tracking coordinate systems. Using both visible and near infrared tracking coordinate systems can enable any one or more of: (a) identifying tools that would not be identified using a single coordinate system; (b) increased pose tracking accuracy; (c) enabling a wider range of motion without losing tracking of surgical instruments, patient anatomy, and/or a robotic end effector; and (d) naturally track an XR headset in the same coordinate system as the navigated surgical instruments.

[0129] FIG. **18** illustrates a block diagram view of the components of a surgical system that include navigation cameras in a pair of XR headsets **1200** and **1210** (head-mounted displays HMD1 and HMD2) and navigation cameras in a camera tracking bar in the camera tracking system component **6'** which houses the computer platform **910**. The computer platform **910** can include the tracking subsystem **830**, the navigation controller **828**, and the XR headset controller **1430** as was earlier shown in FIG. **14**.

[0130] Referring to the surgical system of FIG. **18**, a surgeon and an assistant are both wearing XR headsets HMD1 **1200** and HMD2 **1210**, respectively, each of which includes navigation cameras that may be configured as shown in FIG. **13**. It is optional for the assistant to wear the XR headset HMD2 **1210**.

[0131] The combination of XR headsets HMD1 **1200** and HMD2 **1210** and the navigation cameras **46** on the auxiliary tracking bar can, in operation with the computer platform **910**, more robustly track the example objects of a patient reference array (R), robotic end effector (E), and surgical tool (T) or instrument. The overlapping views from different perspectives that are provided by the XR headsets HMD1 **1200** and HMD2 **1210** and the navigation cameras **46** on the auxiliary tracking bar are shown in FIG. **12**.

[0132] Each of the items labeled in FIG. **18** represent a unique coordinate system. Descriptions of the coordinate system labels are as follows: [0133] A=visible light coordinate system of second headset HMD2 **1210**; [0134] N3=NIR coordinate system of second headset HMD2 **1210**; [0135] S=visible light coordinate system of primary headset HMD1 **1200**; [0136] N2=NIR coordinate system of the primary headset HMD1 **1200**; [0137] N=NIR coordinate system of the auxiliary navigation bar **46**; [0138] V=visible light coordinate system of the auxiliary navigation bar **46**;

[0139] R=NIR coordinate system of a patient reference fiducial array **602**; [0140] T=NIR coordinate system of a tracked tool **604**; [0141] E=NIR coordinate system of a tracked robot end effector on robotic arm **20**; and [0142] W=Inertially navigated world coordinate system with stable gravity vector.

[0143] The spatial relationships of some of these labeled objects (and by extension, coordinate systems) can be measured and calibrated during the manufacturing process, when the equipment is installed in an operating room, and/or before a surgical procedure is to be performed. In the disclosed system, the following coordinate systems are calibrated: T.sub.N2.sup.S; T.sub.N3.sup.A; T.sub.N.sup.V, where the term “T” is defined as a six degree-of-freedom (6 DOF) homogeneous transformation between the two indicated coordinates systems. Thus, for example, the term T.sub.N2.sup.S is a 6 DOF homogeneous transformation between the visible light coordinate system of the primary headset HMD1 **1200** and the NIR coordinate system of the primary headset HMD1 **1200**.

[0144] In one embodiment, the XR headsets HMD1 **1200** and HMD2 **1210** have passive visible light markers painted or otherwise attached to them (coordinate systems S and A), such as the DRA fiducials **1310** shown in FIG. **13**. The navigation cameras are spatially calibrated to these passive fiducials (coordinate systems N2 and N3).

[0145] As explained above, the cameras on the XR headset HMD1 **1200** and HMD2 **1210** and the navigation cameras **46** on the auxiliary tracking bar have partially overlapping field of views. If one or more of the cameras on the XR headset HMD1 **1200** are obstructed from viewing a DRA attached to a tracked object, e.g., a tracked tool (T), but the DRA is in view of the cameras of the other XR headset HMD2 **1210** and/or the navigation cameras **46** on the auxiliary tracking bar, the computer platform **910** can continue to track the DRA seamlessly without loss of navigation. Additionally, if there is partial occlusion of the DRA from the perspective of the cameras on the XR headset HMD1 **1200**, but the entire DRA is visible via cameras of the other XR headset HMD2 **1210** and/or the navigation cameras **46** on the auxiliary tracking bar, the tracking inputs of the cameras can be merged to continue navigation of the DRA.

[0146] More particularly, the various coordinate systems can be chained together by virtue of independent observations the various camera systems provided by the XR headsets HMD1 **1200** and HMD2 **1210** and the navigation cameras **46** on the auxiliary tracking bar. For example, each of the XR headsets HMD1 **1200** and HMD2 **1210** may require virtual augmentation of the robotic end effector (E). While one XR headset HMD1 **1200** (N2) and the navigation cameras **46** on the auxiliary tracking bar (N) are able to see (E), perhaps the other XR headset HMD2 **1210** (N3) cannot. The location of (E) with respect to (N3) can still be computed via one of several different operational methods. Operations according to one embodiment performing chaining of poses from a patient reference (R). If the patient reference (R) is seen by (N3) and either one of (N) or (N2), the pose of (E) with respect to (N3) can be solved directly by either one of the following two equations:

$$T_{N3}^{sup.E} = T_{N2}^{sup.E} T_{N2}^{sup.R} \text{—or—} \\ T_{N3}^{sup.E} = T_N^{sup.E} T_N^{sup.R}$$

[0147] The key to this pose chaining is that the relationship between the frames at the end of each chain are inferred (circled and transported below). The chains can be arbitrarily long and are enabled by having more than one stereo camera system (e.g., N, N2, N3).

[0148] FIG. **21** is a flowchart of corresponding operations that can be performed by a camera tracking system in accordance with some embodiments. Referring to FIG. **21**, the camera tracking system can be configured to receive 2100 tracking information related to tracked objects from a first tracking camera (e.g., N3) and a second tracking camera (e.g., N2) during a surgical procedure. The camera tracking system can determine **2102** a first pose transform (e.g., T.sub.N3.sup.R) between a first object (e.g., R) coordinate system and the first tracking camera

(e.g., N3) coordinate system based on first object tracking information from the first tracking camera (e.g., N3) which indicates pose of the first object (e.g., R). The camera tracking system can determine **2104** a second pose transform (e.g., $T_{sub.R.sup.N2}$) between the first object (e.g., R) coordinate system and the second tracking camera (e.g., N2) coordinate system based on first object tracking information from the second tracking camera (e.g., N2) which indicates pose of the first object (e.g., R). The camera tracking system can determine **2106** a third pose transform (e.g., $T_{sub.N2.sup.E}$) between a second object (e.g., E) coordinate system and the second tracking camera (e.g., N2) coordinate system based on second object tracking information from the second tracking camera (e.g., N2) which indicates pose of the second object (e.g., E). The camera tracking system can determine **2108** a fourth pose transform (e.g., $T_{sub.N3.sup.E}$) between the second object (e.g., E) coordinate system and the first tracking camera (e.g., N3) coordinate system based on combining the first, second, and third pose transforms.

[0149] In some further embodiments, the camera system can further determine **2110** pose of the second object (e.g., E) and the first tracking camera system (e.g., N3) coordinate system based on processing the tracking information through the fourth pose transform.

[0150] Because of the overlapping field of views of the various camera systems, the camera tracking system is capable of determining the pose of the second object (e.g., E) relative to first tracking camera (e.g., N3) when the first camera is blocked from seeing the second object (e.g., E). For example, in some embodiments the camera tracking system is further configured to determine the fourth pose transform (e.g., $T_{sub.N3E}$) between the second object (e.g., E) coordinate system and the first tracking camera (e.g., N3) coordinate system without use of any tracking information from the first tracking camera (e.g., N3) indicating pose of the second object (e.g., E).

[0151] The camera tracking system can be further configured to determine pose of the second object (e.g., E) in the first tracking camera (e.g., N3) coordinate system based on processing through the fourth pose transform the tracking information from the first tracking camera (e.g., N3) which indicates pose of the first object (e.g., R), based on processing through the fourth pose transform (e.g., $T_{sub.N3.sup.E}$) the tracking information from the second tracking camera (e.g., N2) which indicates pose of the first object (e.g., R), and based on processing through the fourth pose transform the tracking information from the second tracking camera (e.g., N2) which indicates pose of the second object (e.g., E).

[0152] The camera tracking system may achieve higher tracking accuracy by merging synchronized imagery from multiple camera systems. For example, the camera tracking system can determine pose of the second object (e.g., E) relative to first tracking camera (e.g., N3) by merging synchronized imagery of the second object (e.g., E) from multiple perspectives (first and second tracking cameras), and can use weighting which can be determined based on accuracy specs of the respective cameras. More particularly, the camera tracking system can be further configured to determine the fourth pose transform (e.g., $T_{sub.N3.sup.E}$) between the second object (e.g., E) coordinate system and the first tracking camera (e.g., N3) coordinate system based on second object tracking information from the first tracking camera (e.g., N3) which indicates pose of the second object (e.g., E) and further based on a result of the combining of the first, second, and third pose transforms.

[0153] The tracking system may be further configured to determine pose of the second object (e.g., E) based on applying a first weight to the first and second object tracking information from the first tracking camera (e.g., N3), applying a second weight to the first and second object tracking information from the second tracking camera (e.g., N2), and processing through the fourth pose transform (e.g., $T_{sub.N3.sup.E}$) the first weighted first and second object tracking information from the first tracking camera and the second weighted first and second object tracking information from the second tracking camera. The value of the first weight indicates pose determination accuracy of the first tracking camera (e.g., N3), and the value of the second weight indicates pose determination accuracy of the second tracking camera (e.g., N2).

[0154] The camera tracking system may be further configured to adapt the value of the first weight over time responsive to determined changes in pose determination accuracy of the first tracking camera (e.g., N3) during operation of the camera tracking system, and to adapt the value of the second weight over time responsive to determined changes in pose determination accuracy of the second tracking camera (e.g., N2) during operation of the camera tracking system.

[0155] The camera tracking system may be configured to: decrease the value of the first weight based on determining the pose determination accuracy of the first tracking camera (e.g., N3) has decreased; and increase the value of the first weight based on determining the pose determination accuracy of the first tracking camera (e.g., N3) has increased.

[0156] The camera tracking system may be further configured to adapt the value of at least one of the first and second weights over time responsive to determined changes in pose determination accuracy of the first tracking camera (e.g., N3) relative to pose determination accuracy of the second tracking camera (e.g., N2) during operation of the camera tracking system.

[0157] The surgical system may be configured to display on the see-through display screen of the XR headset an XR image having a pose that is determined based on the fourth pose transform. The camera tracking system may be further configured to generate the XR image as a graphical representation of the second object (e.g., E) that is posed on the see-through display screen based on processing through the fourth pose transform the first object tracking information from the first and second tracking cameras and the second object tracking information from the second tracking camera.

[0158] As explained above, the camera tracking system can include a navigation controller **828** communicatively connected to the first tracking camera (e.g., N3) and the second tracking camera (e.g., N2) to receive the tracking information and configured to perform the determination of the first, second, third, and fourth pose transforms.

[0159] FIG. **19** show another scenario in which an occlusion prevents cameras on both the XR headsets from directly seeing all of the DRA fiducials attached to the end effector (E), and related pose chaining operations that can be performed to fully resolve the pose of the end effector (E) relative to another tracked DRA (e.g., R). Even if a navigated instrument (e.g., T) or end effector (E) is not seen or recognized by any of the XR headsets' cameras, the navigated instrument (e.g., T) or end effector (E) can still be identified and localized by fusing location information obtained from tracking observable DRA fiducials from multiple perspectives relative to one joint coordinate system (e.g., R).

[0160] When an XR headset is not fully identifying any objects in near infrared or visible light, depending upon the camera configuration, but does see one or more stray DRA fiducials, the tracking light systems (V, S or A) can be used to know roughly where to expect to locate the one or more stray DRA fiducials their respective images. With this added tracking information, individual DRA fiducials can be identified and properly labeled inside of small regions of interest (ROI). The individual stray DRA fiducial information captured from different perspectives can then be combined and correlated in such a way that previously unidentifiable DRAs can now be identified.

[0161] Referring to the example scenario in FIG. **19**, the XR headset **1200** (coordinate system N2) can observe all of the fiducials of the patient reference DRA (R) and can partially observe end effector (E) DRA fiducials (a) and (b) but cannot observe fiducials (c) and (d) because of an intervening occlusive surface **1900**. The other XR headset **1210** (coordinate system N3) can also observe all of the fiducials of the patient reference DRA (R) and can partially observe end effector (E) DRA fiducials (c) and (d) but cannot observe fiducials (a) and (b) because of the intervening occlusive surface **1900**.

[0162] Identifying and solving for the pose of the end effector (E) can be achieved from tracking cameras (N2) and (N3) by using (R) to chain the data into the same frame of reference. Even if neither headset camera system (N2) or (N3) perceive enough DRA fiducials to identify the end effector (E), the four individual DRA fiducials (a, b, c, d) that make up the end effector (E) are

known relative to one another as well as the reference (R) due to pose chaining. What is needed for the end effector (E) to be usefully navigated in surgery is the relationship between the end effector (E) and the patient reference (R). That information can be independently computed as follows from either camera system:

$T_{sub.R.sup.E} = T_{sub.N2.sup.E} T_{sub.R.sup.N2}$ —or— $T_{sub.R.sup.E} = T_{sub.N3.sup.E} T_{sub.R.sup.N3}$
[0163] The pose chaining determines $T_{sub.E.sup.N2}$ and $T_{sub.E.sup.N3}$. In order to detect and solve for the pose of the end effector (E) from the cameras (N2 and N3), the points (a, b, c, d) are put into the same coordinate system. The following translation vector operations can be performed for each stray DRA fiducial found. This provides each DRA fiducial relative to the same patient reference (R):

$$[00001] p_{aR} = p_{aN2} + p_{N2R} p_{bR} = p_{bN2} + p_{N2R} p_{cR} = p_{cN3} + p_{N3R} p_{dR} = p_{dN3} + p_{N3R}$$

[0164] Once the location of all stray DRA fiducials relative to a patient reference are known ($P_{sub.aR}$, $P_{sub.bR}$, $P_{sub.cR}$, $P_{sub.dR}$), a conventional 3D “point to object” detection and pose recovery algorithm can be performed to determine the position and orientation of the robotic end effector (E) relative to the patient reference object (R). Operations of the detection and pose recovery algorithm can include: obtaining a predefined model of the pattern of the DRA fiducials including defined segment lengths between the fiducials; measuring the segments lengths between camera observed ones of the fiducials; and searching and comparing the observed relative segments lengths to pre-defined relative segments lengths from the predefined model of the pattern of the DRA fiducials.

[0165] Various corresponding operations that can be performed by the camera tracking system in accordance with some embodiments are explained below. As explained above, the DRAs can each include spaced apart fiducials which are attached to each of the first object (e.g., R), the second object (e.g., E), the first tracking camera (e.g., N3), and the second tracking camera (e.g., N2). The camera tracking system is further configured, while a first partially hidden fiducial (e.g., a) of the dynamic reference array attached to the second object (e.g., E) can be viewed by the second tracking camera (e.g., N2) but cannot be viewed by the first tracking camera (e.g., N3), to determine pose of the first partially hidden fiducial relative to the fiducials attached to the first object (e.g., R) based on a combination of a determined pose of the first partially hidden fiducial (e.g., a) relative to the fiducials attached to the second tracking camera (e.g., N2) and a determined pose of the fiducials attached to the second tracking camera (e.g., N2) relative to the fiducials attached to the first object (e.g., R).

[0166] The camera tracking system may be further configured, while a second partially hidden fiducial (e.g., b) of the dynamic reference array attached to the second object (e.g., E) can be viewed by the second tracking camera (e.g., N2) but cannot be viewed by the first tracking camera (e.g., N3), to determine pose of the second partially hidden fiducial (e.g., b) relative to the fiducials attached to the first object (e.g., R) based on a combination of a determined pose of the second partially hidden fiducial (e.g., b) relative to the fiducials attached to the second tracking camera (e.g., N2) and a determined pose of the fiducials attached to the second tracking camera (e.g., N2) relative to the fiducials attached to the first object (e.g., R).

[0167] The camera tracking system may be further configured, while a third partially hidden fiducial (e.g., c) of the dynamic reference array attached to the second object (e.g., E) cannot be viewed by the second tracking camera (e.g., N2) but can be viewed by the first tracking camera (e.g., N3), to determine pose of the third partially hidden fiducial (e.g., c) relative to the fiducials attached to the first object (e.g., R) based on a combination of a determined pose of the third partially hidden fiducial (e.g., c) relative to the fiducials attached to the first tracking camera (e.g., N3) and a determined pose of the fiducials attached to the first tracking camera (e.g., N3) relative to the fiducials attached to the first object (e.g., R).

[0168] The camera tracking system may be further configured to process the determined pose of

the first partially hidden fiducial relative to the fiducials attached to the first object (e.g., R) and to process the determined pose of the third partially hidden fiducial (e.g., c) relative to the fiducials attached to the first object (e.g., R) through a pose recovery operation to determine pose of the second object (e.g., E) relative to the first object (e.g., R).

[0169] The camera tracking system may be further configured, while a fourth partially hidden fiducial (e.g., d) of the dynamic reference array attached to the second object (e.g., E) cannot be viewed by the second tracking camera (e.g., N2) but can be viewed by the first tracking camera (e.g., N3), to determine pose of the fourth partially hidden fiducial (e.g., d) relative to the fiducials attached to the first object (e.g., R) based on a combination of a determined pose of the fourth partially hidden fiducial (e.g., d) relative to the fiducials attached to the first tracking camera (e.g., N3) and a determined pose of the fiducials attached to the first tracking camera (e.g., N3) relative to the fiducials attached to the first object (e.g., R).

[0170] Potential advantages that may be provided by various embodiments disclosed herein may include any one or more of: tools that in some scenarios could not be previously tracked can now be robustly tracked by the sharing of partial obstructed views in the same coordinate system; accuracy of head and instrument tracking is improved by merging synchronized imagery from multiple perspectives; a much wider range of motion (position and orientation) is supported without losing tracking of instruments and robotic end effectors due to the expanded overall tracking volume; XR headsets can be tracked naturally in the same coordinate system as navigated instruments without any additional setup; and surgical safety and outcomes are improved due to the built-in tracking redundancy allowing for reliable identification of stereo cameras that are out of calibration or otherwise defective.

[0171] FIG. 20 illustrates a block diagram of information flows between various components of a surgical system operating in accordance with some embodiments of the present disclosure. Referring to FIG. 20, object tracking information flows from the XR headsets **1200** and **1200** to the pose transform chaining module **2000**, and from the navigation cameras **46** on the auxiliary tracking bar to the pose transform chaining module **2000**. The pose transform chaining module **2000** is configured to perform pose chaining in accordance with various embodiments explained above. The XR headset controller **1430** uses the object pose is determined by the pose transform chaining module **2000** to generate XR images that can be provided to the XR headsets **1200** and **1200** for display. The XR headset controller **1430** may generate the XR images for display on the XR headset **1200** based on present pose of the XR headset **1200**, and similarly may generate the XR images for display on the XR headset **1210** based on present pose of the XR headset **1210**. For example, as shown in FIG. 16 the XR headset **1200** can be provided a computer generated graphical representation **1600** of a tool **1602** that is scaled and posed for display on a see-through display screen so the user can accurately manipulate the tool **1602** while viewing the graphical representation **1600** relative to another graphical representation **1610** of an anatomical structure that is to be operated on using the tool during a surgical procedure.

Further Definitions and Embodiments

[0172] In the above-description of various embodiments of present inventive concepts, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of present inventive concepts. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which present inventive concepts belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense expressly so defined herein.

[0173] When an element is referred to as being “connected”, “coupled”, “responsive”, or variants thereof to another element, it can be directly connected, coupled, or responsive to the other element

or intervening elements may be present. In contrast, when an element is referred to as being “directly connected”, “directly coupled”, “directly responsive”, or variants thereof to another element, there are no intervening elements present. Like numbers refer to like elements throughout. Furthermore, “coupled”, “connected”, “responsive”, or variants thereof as used herein may include wirelessly coupled, connected, or responsive. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Well-known functions or constructions may not be described in detail for brevity and/or clarity. The term “and/or” includes any and all combinations of one or more of the associated listed items.

[0174] It will be understood that although the terms first, second, third, etc. may be used herein to describe various elements/operations, these elements/operations should not be limited by these terms. These terms are only used to distinguish one element/operation from another element/operation. Thus, a first element/operation in some embodiments could be termed a second element/operation in other embodiments without departing from the teachings of present inventive concepts. The same reference numerals or the same reference designators denote the same or similar elements throughout the specification.

[0175] As used herein, the terms “comprise”, “comprising”, “comprises”, “include”, “including”, “includes”, “have”, “has”, “having”, or variants thereof are open-ended, and include one or more stated features, integers, elements, steps, components or functions but does not preclude the presence or addition of one or more other features, integers, elements, steps, components, functions or groups thereof. Furthermore, as used herein, the common abbreviation “e.g.”, which derives from the Latin phrase “*exempli gratia*,” may be used to introduce or specify a general example or examples of a previously mentioned item, and is not intended to be limiting of such item. The common abbreviation “i.e.”, which derives from the Latin phrase “*id est*,” may be used to specify a particular item from a more general recitation.

[0176] Example embodiments are described herein with reference to block diagrams and/or flowchart illustrations of computer-implemented methods, apparatus (systems and/or devices) and/or computer program products. It is understood that a block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by computer program instructions that are performed by one or more computer circuits. These computer program instructions may be provided to a processor circuit of a general purpose computer circuit, special purpose computer circuit, and/or other programmable data processing circuit to produce a machine, such that the instructions, which execute via the processor of the computer and/or other programmable data processing apparatus, transform and control transistors, values stored in memory locations, and other hardware components within such circuitry to implement the functions/acts specified in the block diagrams and/or flowchart block or blocks, and thereby create means (functionality) and/or structure for implementing the functions/acts specified in the block diagrams and/or flowchart block(s).

[0177] These computer program instructions may also be stored in a tangible computer-readable medium that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instructions which implement the functions/acts specified in the block diagrams and/or flowchart block or blocks. Accordingly, embodiments of present inventive concepts may be embodied in hardware and/or in software (including firmware, resident software, micro-code, etc.) that runs on a processor such as a digital signal processor, which may collectively be referred to as “circuitry,” “a module” or variants thereof.

[0178] It should also be noted that in some alternate implementations, the functions/acts noted in the blocks may occur out of the order noted in the flowcharts. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Moreover, the

functionality of a given block of the flowcharts and/or block diagrams may be separated into multiple blocks and/or the functionality of two or more blocks of the flowcharts and/or block diagrams may be at least partially integrated. Finally, other blocks may be added/inserted between the blocks that are illustrated, and/or blocks/operations may be omitted without departing from the scope of inventive concepts. Moreover, although some of the diagrams include arrows on communication paths to show a primary direction of communication, it is to be understood that communication may occur in the opposite direction to the depicted arrows.

[0179] Many variations and modifications can be made to the embodiments without substantially departing from the principles of the present inventive concepts. All such variations and modifications are intended to be included herein within the scope of present inventive concepts. Accordingly, the above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended examples of embodiments are intended to cover all such modifications, enhancements, and other embodiments, which fall within the spirit and scope of present inventive concepts. Thus, to the maximum extent allowed by law, the scope of present inventive concepts are to be determined by the broadest permissible interpretation of the present disclosure including the following examples of embodiments and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A surgical system incorporating an extended reality (XR) headset comprising: navigation cameras configured to be attached to a stand or on a secondary headset; an XR headset adapted to be worn by a user, the XR headset having tracking cameras and tracking markers for tracking by the navigation cameras; a patient reference array adapted to be attached to a patient for tracking by the navigation cameras and the tracking cameras; a tracking array attached to an object to be tracked by the XR headset and the navigation cameras; a pose transform chaining module configured to chain pose transforms of two or more of the following: a pose transform between the object tracking array and the XR headset tracking cameras; a pose transform between the XR headset tracking cameras and the patient reference array; and a pose transform between the navigation cameras and the patient reference array; and a pose transform between the object tracking array and the navigation cameras.
